

FROM PHISHING TO DEEPPAKES: EVALUATING INDIA'S PREPAREDNESS AGAINST EMERGING CYBER THREATS

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Abstract

India has always been known for its rapid growth in every aspect, especially technology. India has witnessed a huge progress in technology. But as it continues to grow, a major concern of cybersecurity has also emerged. Earlier, there used to be basic phishing attacks, but now highly sophisticated attacks are taking place, powered by new technologies in India, which are difficult to detect. This paper critically explores the spectrum of these emerging cyber threats and critically evaluates whether India's legal and institutional framework is ready to tackle these threats. It examines India's current legislation on cybersecurity including Information Technology Act, 2000, Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021, Digital Personal Data Protection Act, 2023, Bharatiya Nyaya Sanhita, 2023, CERT In 2013 Rules, 2022 Directions and newly introduced IT (Intermediary Guidelines & Digital Media Ethics Code) Amendment Rules, 2026 along with efforts made by institutions like CERT-In, NCIIPC, and NCCC. The paper also highlights these cyberattacks through a combination of doctrinal analysis, which includes case studies and global comparative analysis. The paper argues that although IT Amendment Rules, 2026 aimed at modernizing India's cybersecurity framework in response to deepfakes, the approach continues to remain reactive and insufficient to address the growing risks posed by deepfakes. The paper also highlights that in order to effectively deal with modern cyber threats, prioritize cyber education and public awareness on these areas. This paper also assesses the India's position in Cybersecurity Ecosystem, offering a comparative analysis with leading international cybersecurity models. At the end of the paper, it has been suggested that India must take a shift from a reactive stance to a proactive cybersecurity governance model. This approach will not only protect the digital infrastructure of India but also preserve the integrity of its democratic institutions and fundamental rights of its citizens in an ever- changing cyber threat environment.

Keywords: *Artificial Intelligence, Cybersecurity, Deepfakes, IT Act, Phishing.*

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INTRODUCTION

A massive rise in rapid digitisation has been seen in the 21st century, which has entirely changed the digital ecosystem of India. India today has a digital population of nearly 850 million, which makes it one of the largest digital ecosystems in the world. This digital surge has been driven by initiatives like Aadhaar Unified Payments Interface (UPI), Digital India and e-governance. But this digital revolution comes with a drawback: the risk of increasing cyberthreats.

A Cyber threat is any potential danger that causes an unlawful activity in computer systems or computing devices, such as smartphones, tablets or networks, with the intention of stealing, disrupting or gaining unauthorised access, such as hacking, phishing or deepfakes. It is often committed by people with a destructive and criminal mindset, either for revenge or monetary gain. Cyber threats can happen in different forms, from using traditional ways such as hacking, phishing, to sophisticated ones such as deepfakes powered by artificial intelligence, which are harder to detect.

Phishing is the most common and traditional form of attacking people digitally. The commonly stolen data are login credentials and credit card numbers. The goal is to send an email or SMS that seems like something the victim needs or wants and induce him/her to click a link. The moment the victim clicks on the link, their personal data gets stolen from them.

Deepfakes are increasingly regarded as modern high-risk cyberweapons, producing AI-generated synthetic videos or audio that impersonate real people to spread misinformation or manipulate public opinion. They seem convincingly real but are actually fake. They blur the line between reality and fiction. Imagine receiving a video message from your CEO asking for an urgent wire transfer. You see their face. You hear their voice. Would you question it? That is the dangerous power of deepfakes; they can trick and manipulate people.

The shift from phishing to deepfakes highlights the increasing complexity of technologies used in cyberattacks, such as machine learning, artificial intelligence and big data analytics. The implications of cyber-attacks can be financial loss, privacy invasion, political manipulation, or even national security threats, which means cybersecurity is no longer a technical issue but a matter of democratic stability and social trust. In India, although there are laws like Information Technology Act, 2000, Digital Personal Data Protection Act, 2023, Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021, Bharatiya Nyaya Sanhita, 2023, the CERT In 2013 Rules and 2022 Directions which provides safeguards and institutions like CERT-In, NCIIPC, and NCCC, but these laws have no explicit legal provisions against

deepfakes. However, the newly introduced IT Amendment Rules, 2026, explicitly address deepfakes and focus on regulating AI-generated content, but continue to remain reactive and insufficient to address the growing risks posed by deepfakes.

OBJECTIVES OF THE STUDY

1. To explore the changing nature of cyberattacks from phishing to deepfakes.
2. To examine the extent to which the current legal framework of India is equipped to deal with modern cyber threats.
3. To analyse gaps in India's enforcement and preparedness for advanced cyber threats.
4. To compare India's cybersecurity ecosystem with the best global models of the US, the EU and China, with an aim of identifying adaptable best practices
5. To suggest a high-priority transition from a largely reactive stance to a proactive future branded cybersecurity governance.

RESEARCH QUESTIONS

1. How has the evolution from traditional cyberattacks to AI-driven threats revealed inadequacies in India's existing legal framework?
2. Does India's existing legal cybersecurity framework adequately address the challenges posed by modern cyberattacks?
3. How can India transform from a reactive to a proactive regulatory model by adopting global best cybersecurity practices?

SCOPE OF THE STUDY

1. **Geographical Boundaries:** The study is not only confined to India but also includes a comparison with leading international cybersecurity models focusing on their legal and technological preparedness against emerging cyber threats.
2. **Subject Scope:** Various types of cyber threats have been covered in the paper. However, the main focus has been given to phishing and deepfakes.
3. **Legal and Institutional Focus:** The paper assesses India's current legal position in tackling cyberattacks. Legislation like the Information Technology Act, 2000 and the Digital Personal Data Protection Act, 2023, etc., has been taken into consideration. The paper also examines the efforts taken by government institutions like CERT-In, NCIIPC, Cyber Crime Units, and NCCC.

4. **Technological Scope:** The paper talks about the different types of technologies behind the cyberattacks, such as AI and machine learning.

METHODOLOGY

1. **Doctrinal Method:** The paper examines India's current legislation dealing with cybersecurity, which includes an in-depth analysis of the Information Technology Act, 2000 and the Digital Personal Data Protection Act, 2023, Bharatiya Nyaya Sanhita, 2023, CERT In 2013 Rules and 2022 Directions and review of institutional efforts taken by the government, such as CERT-In, NCIIPC, Cyber Crime Units, and NCCC.
2. **Comparative Analysis:** International Cybersecurity Models Assessment of advanced international cybersecurity models from countries such as the United States, the European Union, and Singapore
3. **Secondary Data Collection:** Use of published research papers, government reports, legal documents, news articles, and official cybersecurity reports for analysis of trends, statistics, and expert opinions to evaluate the effectiveness of India's cybersecurity laws

LITERATURE REVIEW

Evolution of Cyberthreat in India

In India, the date dates back to the initial stages of internet usage, when the nation first started witnessing the onset of digital fraud. Studies show that one of the earliest notable incidents happened in the mid-1990s, when a group of hackers impersonated AOL employees through instant messaging and email to steal users' passwords and hijack their accounts.¹ These messages used to request login credentials for account verification. This era established the basic framework of phishing, using fraudulent communications to trick users into handing over sensitive information. However, it is not extensively documented in academia.

Another incident was a PayPal scam in the early 2000s, where attackers impersonated PayPal under names like "PayPai", which targets account holders of the widely used internet payment service PayPal, taking advantage of the fact that a capital "i" may be difficult to distinguish from a lower-case "l" in some computer fonts. The scam involves sending PayPal account holders a

¹ J Crumbaugh, "The evolution of phishing attacks", *available at*: <https://www.phishfirewall.com/phishing-playbook-chapters/the-evolution-of-phishing-attacks> (last visited on: 3.04.2026).

notification email claiming that PayPal has “temporarily suspended” their account. Instead of linking to PayPal.com, the site references in the email link to a convincing duplicate of the site at paypai.com, in the hope that the user will enter their PayPal login details, which the owner of paypai.com can then store and use.²

By the early 2010s, phishing had evolved from small-scale fraud to organised digital networks, as in the case of the Jamtara scam in Jharkhand, including the tools and techniques employed by offenders, as well as the socio-economic factors that contribute to their activities. In India, in this case, the scammers used to call people, pretending to be bank officials or telecom executives. They created fabricated situations, like threatening to freeze bank accounts or disable SIM cards, to create a sense of haste. Targets were then bullied into divulging sensitive data, such as bank account numbers, OTPs, and ATM PINs. Over time, the operation expanded, with numerous youths in Jamtara replicating the scheme, earning the district the reputation of India’s “phishing capital.”³

The COVID-19 pandemic marked another surge in phishing attacks, as cybercriminals exploited public fear and the increased reliance on digital services. A study noted that during the Covid 19 pandemic, there was a surge in phishing attacks. Because of the lockdown, people were heavily relying on digital services. Cybercriminals took advantage of this situation by sending malicious emails in the name of charitable purposes to the victims who suffered during the pandemic.⁴

Legal and Institutional Framework in India

A study showed that India has undergone significant development in terms of cybersecurity laws in order to keep pace with emerging digital risks, but is still not fully equipped to tackle emerging cyberthreats like deepfakes. No doubt, the IT Act 2000 acts as the cornerstone of India’s legislation on cybersecurity and has been supported by several subordinate rules aimed at tackling cyber-related risks. Though India has expanded its cybersecurity regime, its effectiveness is significantly affected by enforcement and regulation challenges. Considering the fact that cybercrime may transcend national borders, India should show its willingness to strengthen international partnerships. Such collaboration will help in the exchange of knowledge and coordinated strategies. Moreover, keeping in mind that technological advancements have

² E Gabrilovich and A Gontmakher, “The homograph attack” 45(2) *Communications of the ACM* 128 (2002).

³ R Suman, P.K. Ray and A Sarkar, “A study on cyber financial frauds in the district of Jamtara, Jharkhand” 9(4) *International Journal of Research and Analytical Reviews* 123 (2022).

⁴ M.L. Kumar and P.S. Akram, “Cybersecurity laws in India: Constitutional implications and regulatory challenges” 14(15s) *Journal of Neonatal Surgery* 193 (2025).

transformed simple cyberattacks into sophisticated ones, India should take a shift from a reactive approach to a proactive approach in order to fight against the emerging modern cyberattacks.⁵

Technological aspects of Cyber Threats and AI's Role:

With the emerging rise of new digital advancements, the nature of cyber threats has also transformed. Earlier, they were easy to detect, but modern cyberattacks powered by AI are very difficult to detect due to their adaptive and intelligent nature. The question of whether AI has strengthened or weakened cybersecurity defences has always been debated. Different scholars have different reasonable opinions about it.

Some studies show that AI enhances cybersecurity defences as it relies on several advanced technologies, including machine learning, deep learning and natural language processing (NLP). Machine learning is a subset of AI that is more focused on the use of various self-learning algorithms that derive knowledge from previous data in order to predict outcomes, which helps in preventing cyberattacks by recognising similar patterns in future. Deep learning, also known as neural networks, is a further subset of machine learning. It teaches a computer with many layers of neural networks to recognise complex patterns given in the data, so that it will learn to see, hear and understand things on its own. Natural language processing (NLP) helps AI to read, understand and derive meaning from human languages.⁶

However, some studies say that the technologies, specifically AI, which boost the cybersecurity defences, are the same that can worsen the landscape of cyberattacks, as AI can now generate fake images, videos or audios that seem difficult to identify by a normal person, expanding the attack surface. Computer Using Agents (CUAs), built on platforms like OpenAI Operator, are now able to mimic human behaviours, which raises serious concerns.⁷

This shows how AI can both strengthen or worsen the cybersecurity defences. The focus should be on using AI for strengthening purposes, which requires both technical implementation and strategic oversight.

Gap identification

⁵ M.L. Kumar and P.S. Akram, "Cyber security laws in India – Constitutional implications & gaps" 14(15S) *Journal of Neonatal Surgery* 194 (2025).

⁶ Syracuse University School of Information Studies, "AI in cybersecurity", *available at*: <https://ischool.syracuse.edu/ai-in-cybersecurity/> (last visited on: 03.04.2026).

⁷ T Seals, "Agentic AI: Cybersecurity's friend or foe?", *TechRadar Pro*, Oct. 31, 2024, *available at*: <https://www.techradar.com/pro/agentic-ai-cybersecuritys-friend-or-foe> (last visited on: 03.04.2026).

- Although India's cybersecurity legislative framework has undergone significant developments, it is still not fully compatible with dealing with advanced cyberattacks and lacks a reactive and preventive approach.
- Socio-economic factors have always been ignored, as most of the studies focus only on the technical aspects of cybercrime.
- There is a lack of comprehensive research on modern cyberattacks like deepfakes and synthetic media.

UNDERSTANDING CYBER THREATS

A cyber threat is any malicious attempt carried out by either an individual or group through the use of digital tools with the aim of gaining unauthorised access, manipulating data, threatening security, privacy or causing damage. It consists of a wide range of attacks such as phishing, deepfakes, identity theft, and synthetic media.

Classification of cyber threats

1. **Financial Crimes:** In the cyber context, it refers to illegal activities that involve the exploitation of banking networks and digital payment systems, with the help of technological tools like phishing and malware, with the aim of cheating people or stealing sensitive financial details, such as Credit or debit card skimming, UPI fraud, and trading app fraud.
2. **Data Theft:** It is the most common form of cyber threat, which includes illegal extraction of personal data by sending fake links or emails, from databases through malware attacks and hacking. Data theft may include banking information, passwords, photos or any other confidential information.
3. **AI-based manipulation:** As a cyberthreat, AI-based manipulation refers to the modern cyberthreats that are powered by AI to deceive or influence individuals through synthetic media, targeted content, and voice cloning. The fact that AI can analyse behaviour and personal data makes it difficult for normal people to distinguish between real and fake content, such as deepfake videos used for blackmailing, social media bots spreading misinformation, etc.

EVOLUTION OF CYBERCRIME - KEY TRENDS, CYBERSECURITY THREATS, AND MITIGATION STRATEGIES FROM HISTORICAL DATA

According to the study, historically, cybercrimes used to be simple in nature, such as sending fake emails, messages or links to trick people. Such kind of scams are easy to detect because they include suspicious sender addresses, threatening language, poor grammar, etc. But in the last few decades, there has been a huge shift from simple cyberattacks to AI-powered cyberattacks due to rapid digitisation. AI-powered cyberattacks are complex in nature, as AI can analyse behaviour and personal data, which is why they are more convincing.⁸

Earlier, the main target of cybercriminals used to be small enterprises and individuals, as they had limited knowledge about the digital world and technology. This trend is not only technological but also closely linked to underlying socio- economic factors that significantly shape the dynamics of cybercrime in India. In India, differences in digital literacy and educational disparities have a strong impact on both victimisation and offender behaviour. On one hand, a digitally unaware population are disproportionately targeted, making them more vulnerable to cyberattacks as they are not capable of identifying them. On the other hand, disadvantaged groups may find cybercrime an attractive alternative due to its low risk and high reward opportunity. This dual dynamic shows that cybercrime is not about only technology but also requires consideration for socio- economic factors. The case of Jamtara shows how socio-economic factors can foster environments conducive to cyber fraud.

As the technology kept improving, so did the cybercriminals. Technological advancements have spurred the growth of complex cybercrime. The total number of ransomware attacks has increased in contemporary times, where cyber criminals illegitimately encrypt victims' information and then extort substantial ransom payments to provide the decryption key. Not just individuals and corporations have also been impacted by these attacks. Advanced persistent threats (APTs) are another notable trend. These attacks tend to be state-sponsored or state-funded; they are prolonged, stealthy penetrations of complex computer systems to capture confidential information or disrupt operations

The landscape of cybercrime has changed over the years, too. In their earlier days, only small-scale businesses and individuals were targeted by cyber criminals due to insufficient security resources and a lack of awareness. However, with every breakthrough in technology, cyber criminals develop more sophisticated plans, owing to which big organisations, such as the private sector, governments, civil society, and the military, are becoming common targets.

⁸ M Abdullah, M.M. Nawaz, B Saleem, M Zahra, E.B. Ashfaq and Z Muhammad, "Evolution cybercrime-Key trends, cybersecurity threats, and mitigation strategies from historical data" 4 (3) *Analytics* 25 (2025).

The early 1960s were the period when early technological developments birthed to connectivity. The computers during this era were costly, big, and huge. Multiple individuals shared a single computer simultaneously. This sharing of time led to the necessity of avoiding access by unauthorised users to computers and their data. And thus, the idea of data security and hardware security was conceived here. The defence practices during this decade were more focused on physical access control, locked facilities and duties segregation to avoid insider abuse. During the 1970s, ARPANET, the initial version of the internet, was established, which provided hackers everywhere with much to ponder. Some early malware, such as the Creeper and Reaper, were created as well, but they were used as academic exercises, not as actual malware. In 1975, an article called 'The Protection of Information in Computer Systems' was released that provided rules that would serve as the basis of contemporary cybersecurity. The 1980s witnessed the formal establishment of the Internet, and the networks everywhere in the world began implementing the Internet Protocol Suite. The adoption added additional prey and foes to the table. The dictionary attack, which is employed to take advantage of weak or default passwords, was also first deployed during the 1980s. Since the world of the internet was new, the security practices to be implemented were also not yet well understood in this context. In the 1990s, the period is referred to as the virus era. As personal computers became common, they enabled even new users to gain access to ready-made hacking tools without understanding coding. No doubt, many of these attacks were done for experimentation or curiosity, but they highlighted growing risks of cyber threats for technological companies, promoting the adoption of antivirus software, mandatory updates, etc. The 2000s saw the growth of online banking, and reliance on digital services increased, which created opportunities for cybercriminals to commit financial fraud. During the 2010s, there was further evolution of cyberattacks with the emergence of state-sponsored attacks, backed by the government, regarded as the most sophisticated form of cyberattacks. These cyberattacks include a high-skilled hacking group that used to target tech giants and global organisations in order to steal information. The 2020s saw a transition from traditional cyberattacks such as phishing to advanced cyberattacks like deepfakes, which are more difficult to detect as they are powered by AI. During this time, cybercriminals started to use fake but realistic audio and video to pretend to be something else and trick people into sending money or revealing their personal information. A systematic review also brings forth continuing challenges such as under-reporting of cybercrime, lack of adoption of sophisticated detection solutions, and incorporation of psychological, social, and technical factors in research designs. Cumulatively, these findings underscore that the 2020s require a multidisciplinary response leveraging AI-driven detection, regulatory reforms, public awareness, and international

cooperation to effectively counteract evolving cybercrime threats. Talking about the present context, as we are entering a more globalised world each passing day, the threats and dangers of cybersecurity are growing as well, like deepfakes. Emerging technological innovations such as 5G, quantum machines, IoT appliances, and cloud computing services have expanded the attack surface. Therefore, in response to these developments, strong and proactive countermeasures and cautionary steps must be applied.

PHISHING

The term “Phishing” has drawn its resemblance from the word “fishing”, in which one uses an insect as a lure to catch a fish. The moment the fish bites the insect, it gets hooked to the bait. Similarly, phishing is being used. In the digital world, to defraud the victims by luring them into a bait. The forms of bait could take place in the form of a phishing email with the URL of a phishing website, or smishing, a type of phishing scam that takes place through SMS, or vishing / voice phishing, another type of phishing scam that takes place via phone calls. By using all these methods, the phisher tricks the users into revealing personal information such as login credentials, Credit Card CVV, biometrics, details of Aadhaar, Gmail IDs, or any other confidential information that seems valuable to him.⁹ Upon falling for the phisher’s trick, the victim enters their confidential details on this phoney website, the phisher then steals this information and misuses it to transfer the money of the victim.

According to studies, there is a high possibility that the phishers sell the stolen data, such as login credentials or any other sensitive information, on the dark web community in exchange for a large amount of money. The community then initiates further such attacks through these compromised machines or servers. The phishers may also equip themselves through the Dark Web, further helping them with these phishing attacks and expanding their operations. These phishers often inculcate more than one method or technique to initiate more and more such attacks and trick victims through emails, SMSs, etc., to click on fake URLs. Once successful, malware is discreetly downloaded into their machines, and thus, complete, uninterrupted, and discreet access to these networks and machines is with the phisher through additional system Administrator Accounts.¹⁰

There was a surge in phishing email attacks during COVID-19. The Economic Times reported that COVID-19-related phishing attacks grew by 667% in February 2020. Phishing attacks were

⁹ Jaya, “Machine learning based phishing website detection” (2023) (Unpublished doctoral thesis, Jawaharlal Nehru University).

¹⁰ A Oest *et.al.*, “Inside a phisher’s mind: Understanding anti-phishing ecosystem through phishing kit analysis”, in *2018 APWG Symposium on Electronic Crime Research (eCrime)* 1 (2018).

happening in the name of donations for COVID-19, help for health workers and free online meeting software. A report publicised by APWG in its 1st Quarter, 2020, stated that some of the email phishing attacks were carried out by sending fake Zoom video-conferencing meeting notifications to people.¹¹ The report by APWG in its 3rd Quarter, 2022 Phishing Activity Trend Report highlighted that the number of unique Phishing Websites detected in the months of July-Sep 2022 was 425,112 (July 2022), 430,141 (Aug 2022), and 415,630 (Sep 2022).¹²

DEEPPFAKES

Deepfake is one of the most advanced cybercrimes that uses artificial intelligence (AI), which allows the user to mimic the synthetically enhanced video of a real person, providing images and audio-visual to make the user speak or behave in a manner that is not realistic. As deepfakes are based on machine learning, data and information of a person can be easily processed by algorithms and systems, thereby generating any body part on screen, including face, body and other visual characteristics, which appears to be real but is not true. There are three main groups that have been identified by the researchers into which deepfake content falls. The first is Face Swap, replaces a person's face in a video or image with someone else's, creating a seamless and natural looking transformation; the second is Lip- sync, manipulates a person's lip movements to match new or altered audio track, making it appear that they are speaking words they never said; and the third is puppet- master, manipulates the facial expressions of one input at a time. The person in the source video controls the facial expression and movements of the person in the target video.¹³

Deepfakes are a subdivision of deep learning, which is a realm of artificial intelligence that depends on neural networks for its operation. In the realm of deep learning, typically a combination of input and output layers is used to operate through neural networks. The system works in gears with the two main components under the framework, named the generator and the discriminator. Both these components are responsible for the algorithm in use. These two bodies help the system to differentiate between synthetic and authentic data, wherein the generator helps in mimicking real content through artificial outputs, and the discriminator helps in identifying the features that point out fabricated material and also helps check authenticity. As these components function and interact with each other, the sync and specific functions allow the generator to continuously receive feedback from the discriminator, which identifies authentic

¹¹ Anti-Phishing Working Group, "Phishing activity trends report: 1st quarter 2020" (May 11, 2020), *available at*: https://docs.apwg.org/reports/apwg_trends_report_q1_2020.pdf (last visited on: 03.04.2026).

¹² Ibid.

¹³ Ibid.

traits detected by it and allows fake material to become more realistic and standardises it with genuine examples. As the distinction between synthetic and authentic images grows smaller, so does the significance of the input that the model assigns to itself, utilising these measures as a metric for performance accuracy. The elaborative process shapes the foundational base for deepfakes and is called Generative Adversarial Network (GAN), which possesses the capacity to generate and reproduce data patterns to generate deepfakes. The discriminator here plays the most important role in keeping generating feedback, which helps the model to correct itself wherever it lacks authenticity. Without the feedback in place, the model would fail and would not be able to correct itself.¹⁴

INDIA'S LEGAL AND INSTITUTIONAL FRAMEWORK ON CYBERSECURITY

India's legal and institutional framework addresses cybersecurity in India by providing privacy and securing critical infrastructure. India's legal, institutional and regulatory framework has expanded steadily, shaped by several laws, statutes, rules, policies and institutional mechanisms. All these instruments aim to define cybercrimes in order to protect personal information, online transactions and overall ensure a safe digital environment for individuals, corporations and the government.

Legal Framework

The foundational legislative framework that deals with cybersecurity in India is the Information Technology Act, 2000, considered the cornerstone of the country's digital legal architecture. It was enacted when the trend of electronic communication and e-commerce was emerging globally. The Act prevents cybercrimes, regulates electronic transactions and legitimises digital signatures. A wide range of offences and responsibilities has been addressed by the Information Technology Act, 2000. Section 43 of the IT Act, 2000 punishes the offender for data theft, identity theft, hacking a computer system, damaging a computer system or computer network, or causing denial of service. Section 43A of the IT Act, 2000 provides compensation to the affected person if a body corporate viz. any company and includes a firm, sole proprietorship or other association of individuals engaged in commercial or professional activities. Every so often, the generator may generate such refined data that the discriminator fails to tell it apart from the actual and real data, creating a loop wherein both parties oscillate around like a pendulum,

¹⁴ J Gurumurthy, "In the pursuance of a robust legal framework to address deepfake harms: An analysis of the Indian legal discourse" 20(1) *Indian Journal of Law and Technology* (2025).

forming an uncertainty. India's Framework on cybersecurity helps address this issue by providing privacy and securing critical infrastructure.¹⁵

Also, the Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021, notified under the Information Technology Act, 2000, establish a new framework for regulating social media companies, online intermediaries, OTT platforms and digital news publishers, to ensure accountability in relation to their activities. These rules make the "safe harbour" protection under section 79 of the IT Act, 2000 contingent on the intermediary exercising due diligence, including publishing user-agreement policies, responding to complaints within 24 hours and taking down unlawful content within 72 hours of a request by a court or government authority.¹⁶ The rules also classify online platforms as Social Media Intermediaries (SSMIs), platform with over 5 million users in India. SSMIs are required to appoint a chief complaint officer, a nodal contact officer, and a resident grievance officer; store user data for 180 days and publish monthly transparency reports.¹⁷ The most controversial aspect of the rules is the traceability requirement, which mandates that encrypted messaging platforms like WhatsApp must make the first originator of the message identifiable for investigations involving certain types of offences.¹⁸ Scholars argue that traceability undermined end to end encryption and disproportionately infringes on the right to privacy recognized in *Justice K.S. Puttaswamy v. Union of India*.¹⁹ For OTT platforms, The Rules implement a three-tier regulatory arrangement whereby there is a level of self-regulation by the platforms, a level of self-regulation by the industry body, and a level of final determination by a government committee, while digital news publishers must follow the Press Council and Cable Television Programme codes.²⁰ While the Rules were designed to address illegal content and misinformation by enhancing accountability, data shows that unclear obligations to remove certain content and state-controlled oversight will likely lead to censorship and create a chilling effect on free speech.²¹

An amendment came in 2008 in the IT Act 2000, which introduced 9 chapters and 117 sections, which updated the law to tackle modern cybercrimes, reinforce security and data protection standards. It also supported the use of digital signatures, strengthened forensic processes and

¹⁵ M Gupta and A Gupta, "Cyber security legal framework in India - Overlaps, problems and challenges" 12(1) *Journal of Business Management & Information Systems* (2025).

¹⁶ The Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021, r. 3.

¹⁷ Ibid at r. 4.

¹⁸ Ibid at r. 4(2).

¹⁹ (2017) 10 SCC 1.

²⁰ Supra note 16 at pt. III, rr. 8-13.

²¹ G Bhatia, "The Indian Intermediary Guidelines: A constitutional analysis", *Indian Constitutional Law and Philosophy*, Feb. 26, 2021, available at: <https://indconlawphil.wordpress.com/2021/02/26/the-indian-intermediary-guidelines-a-constitutional-analysis/> (last visited on: 03.04.2026).

cybersecurity investigation and acknowledged the necessity of protecting personal data from cyberthreats. Also, the mandate imposed on organisations and intermediaries to notify CERT-In of cybersecurity incidents involves a more structured approach to strengthen the responsive cybersecurity framework.

India has strengthened its legal framework through the Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011, also known as Privacy rules, which were enacted as an accessory to the IT Act, 2000. They recognised certain categories of information, such as biometric data, financial details, passwords, and medical data, as Sensitive Personal Data or Information (SPDI). It imposed compliance requirements on intermediaries and other such entities handling such data. Companies and intermediaries were required to obtain consent from the user, ensuring transparency regarding data usage and implementation of reasonable security practices to prevent unauthorised access. The rules also mandated the establishment of grievance redressal mechanisms and obligated the publication of privacy policies. Later, influenced by global data protection frameworks like the EU'S GDPR, India enacted a comprehensive personal data protection law on August 11, 2023. This law aims to create a secure and accountable data processing environment by allowing individuals to access, correct and withdraw consent for the processing of their personal information. The Act also imposes reasonable yet strict restrictions and obligations upon the Data Fiduciaries (organisations dealing with data) in relation to the regulations and limitations of their power over personal data, and most importantly, to immediately notify any breach to their stakeholders, ultimately reinforcing compliance maintenance and security safeguards. The issue of the protection of children's data has also been taken into account while making the DPDP Act, as the Act ensures that minors are not subject to targeted advertising and data profiling of minors. To oversee and ensure proper and efficient implementation of the DPDP Act, a quasi-judicial body has been set up called the Data Protection Board (DPB) by virtue of the Act. The nature of the Board being that of a quasi-judicial one, it is empowered to not only investigate any violations but also impose penalties on such violators as required and prescribed. The newly enacted DPDP, as mentioned above, is not one of a limited nature offering limited regulations, but is an overarching act that helps in almost every aspect. The act ranges from establishing a safe and secure environment for encouraging business, boosting the economy, by adopting measures for accountability, trust, and compliance. It also addresses national security concerns relating to cyber-attacks and empowers the state and central government to process data for national security. The impact from these establishing and first time legislations cyber security awareness has reached a new level in India and key legislations such the penal code of India now

Bhartiya Nyaya Sanhita, 2023 (BNS) under section 2(39) of the act to increase compatibility and application of the IT act mentions that words and meanings in relation with electronic and digital means is inculcated same as it is in the IT act, thus helping in deterrence and detection crimes through such means. These clarifications and inclusions now help ensure that the core and traditional criminal law is not unknown to the modern technological upgrades, but complements the specialised IT Act 2000.

In 2013, a subordinate legislation aiming to improve cybersecurity laws was enacted, known as the Computer Emergency Response Team (CERT-In) Rules. Through this act, CERT-In was designated as the nodal agency for monitoring and reporting any incident, all the while providing quick response to these incidents. It is empowered to collect data anytime an incident takes place, issue advisories and also seek information from intermediaries, data centres, and service providers to prevent any threats in a timely manner. To add more safeguards and checks, these rules were updated in 2022, which added more obligations such as incident reporting within six hours, retaining data of users for a maximum of 180 days, VPN firms, virtual assistance and much more. These guidelines are foundational grounds for a step towards proactive surveillance and accountability, all the while matching itself with global standards.

A major shift comes with the IT (Intermediary Guidelines & Digital Media Ethics Code) Amendment Rules, 2026. The new rules come into effect from 20 February, 2026. This amendment makes a turning point in India's legal framework governing cybercrime, as it is the first framework to explain the concept of "Synthetically Generated Information" (SGI) explicitly under Rule 2(1)(wa). It defines SGI as AI-generated or AI-altered content which appears to be real, and can mislead users into believing it is authentic, like deepfakes, AI-generated realistic images, and voice cloning. The amendment mainly focuses on regulating AI-generated content. Not every AI-generated content is SGI, like basic editing, which includes noise reduction, video compression, adding subtitles, and normal AI assistance for creating PowerPoint slides, drafting samples. Therefore, this amendment differentiates between content that SGI and content that does not.²²

Key features of the IT (Intermediary Guidelines & Digital Media Ethics Code) Amendment Rules, 2026:

²² Prime Infoserv, "AI & Deepfake Law in India: IT Rules 2026 Amendment Tightens Control Over AI-Generated Content", *Prime Infoserv*, Feb. 11, 2026, available at: <https://primeinfoserv.com/blog-ai-deepfake-law-india-it-rules-2026-amendment/> (last visited on: 03.04.2026).

1. Reduction in response time:

- Intermediaries have to take down the AI-generated content within 3 hours upon receiving court or government orders. Earlier, the time period was 36 hours.
- The timeline for resolving general complaints has been reduced from 15 days to 7 days.
- The timeline for removal of morphed intimate images/ nudity has also been reduced from 24 hours to 2 hours.

2. Mandatory labelling of SGI content:

- Under Rule 3(3), the intermediaries have a mandatory duty to systematically label the SGI content on their platform.
- It must display permanent and unique metadata.
- For deepfake videos, the label must occupy 10% of the screen area and must remain permanently visible.
- For audio deepfakes, mandatory audible disclosure during the opening seconds enables viewers to evaluate the credibility of the content.

3. SSMI (Significant Social Media Intermediaries) requirements:

- Intermediaries with more than 5 million users are considered SSMI.
- SSIMIs are required to deploy AI tools or human review to verify user declarations. The new rule 3(1A) gives protection to intermediaries if they remove synthetic content honestly, either through reasonable efforts or acting on user complaints.

4. Mandatory warning for using AI tools:

- If a platform is providing AI tools to use, then it must also inform the users about the legal penalties for misusing the same
- They must regularly show warnings at sign-up and periodically to ensure continued awareness.

Institutional Framework:

With all these legislations in place, it is clear that cybersecurity governance has taken a step forward; however, these legislations only bind the provisions within the legal boundaries, and thus, the role of implementation further enhances the need for well-structured, accountable, and transparent institutions. India has several dedicated institutions aiming at the prevention of cybercrimes through reporting, policy execution, and mitigation. One of the key institutions in

this context is the Computer Emergency Response Team (CERT-In), which was established in 2004 and plays a crucial role in safeguarding the digital infrastructure of India. It enables India to collaborate with national and international organisations in conducting awareness campaigns. Through its secure and complex network of intelligence and coordination with the cybersecurity framework, the institution has now come to play an indispensable role in maintaining the integrity and security of the nation's digital ecosystem. On 16 January 2014, the National Critical Information Infrastructure Protection Centre (NCIIPC) was officially notified in the Information Technology (Amendment) Act, 2008, under Section 70A. This is another initiative taken by the government. This body is the supreme body for securing Critical Information Infrastructure (CII). NCIIPC is a subset of the National Technical Research Organisation (NTRO) and performs functions such as identifying cybersecurity vulnerabilities across the critical sectors like power, telecom, transport, finance, and government) performing risk assessment and audit, and sharing actionable threat intelligence with stakeholders in the relevant sectors. Through its strong mandate, it focuses on prevention and makes sure that the national sovereignty and functioning are not affected by a cyberattack on critical infrastructure.

Ministry of Electronics and Information Technology (MeitY) has also implemented an institution that helps in real-time analysis and threat prediction called the National Cyber Coordination Centre (NCCC). Its role is to analyse any internet traffic anomalies, send early alerts, and distribute intelligence.

THE DILEMMA OF INEFFICIENT INDIAN LAWS

Though the IT Amendment Rules, 2026 make a significant step forward in India's approach to regulate deepfakes, they are not sufficient on their own. The major drawback is that these amendments are intermediary-centric. It tells platforms to regulate the content, detect harmful AI-generated content, label it and take it down if it misled viewers. However, it does not directly regulate the creation or misuse of deepfakes themselves in India. There is still no specific law defining deepfake offences, no direct punishment for creators, and no specific law targeting its misuse. Therefore, these amendments govern the platforms where harmful AI-generated content is shared rather than the act of creating harmful AI content. These new rules also ignore the ground reality that harm goes beyond platforms. Even if intermediaries are well-regulated, such content can still be shared in private. For example, apps like Telegram and WhatsApp have encryption that makes it difficult to detect such content. Also, people can share such content by sharing files directly from one device to another via Airdrop, Bluetooth or USB services without involving any platform. It helps in regulating the platform, which is only one part of the

problem, but fails to regulate the harm itself. Therefore, the law is not fully effective and continues to remain reactive as it does not address the actual misuse of deepfake technology.

Another major gap left unaddressed by India's cybersecurity legislative framework is jurisdiction. This is mostly true for cross-border crimes, wherein a variation between the jurisdictional gap creates a grey area for cybercriminals, giving them opportunities to escape from liability by operating from within nations or places that are heavily ill-equipped with their cybercrime laws, leading to limited enforceability. Such variability is often seen in these cyber-attacks, mostly prominent for hacking. Hacking often is initiated from countries with outdated cyberlaw, making it difficult for law enforcement to not just apprehend criminals but also help in mutual legal compliance with other countries. Additionally, advancements in hacking have allowed it to start the same attack from not one but several jurisdictions, making it even tougher to trace and locate the origins of the hacker. Thus, such an inconsistency in the legal framework often leads to a lack in the justice system when it comes to a speedy response to such attacks, as law enforcement often is ambiguous or stuck in its own procedures, protocols, and timing.²³ Studies also suggest that India's lack of action against transboundary crimes is not in its framework, but rather in the related issues of its ability to collect, preserve, and admit evidence outside the jurisdiction of Indian boundaries.

Various policy papers have suggested that India is very much dependent upon the Mutual Legal Assistance Treaty (MLAT), an official agreement between two or more nations that allows signatories to exchange evidence and information for purposes of investigation and prosecution of crime, which is unstandardized and slow and results in the loss of important server log evidence.²⁴ To address these issues, scholars have put forth a response model that is in 3 layers. Firstly, at the legal level, India should enter into bilateral data access treaties with significant jurisdictions similar to the U.S. CLOUD Act and implement emergency data preservation orders with the ability to bind foreign intermediaries.²⁵ Secondly, at the institutional level, India should designate the NCCC as the key agency to coordinate CERT In, NCIIPC and the 24/7 Interpol Network for cross-border evidence.²⁶ Third, at the diplomatic level, researchers have suggested

²³ M.T. Hasan, "Cross-border cybercrimes and international law: Challenges in ensuring justice in a digitally connected world" 4(1) *IJRDO Journal of Law and Cyber Crime* 1–7 (2024).

²⁴ Carnegie India, "Cross-border data access for law enforcement in India", *Carnegie Endowment for International Peace*, June 12, 2020, available at: <https://carnegieindia.org/2020/06/12/cross-border-data-access-for-law-enforcement-in-india-pub-81981> (last visited on: 03.04.2026).

²⁵ SSRN, "Issues with enforcing MLATs in cybercrime cases: An empirical study" (2022), available at: <https://ssrn.com> (last visited on: 03.04.2026).

²⁶ CCDCOE (Cooperative Cyber Defence Centre of Excellence), "Mutual legal assistance and improving international cooperation in cybercrime investigations" (2021), available at: <https://ccdcoe.org> (last visited on: 03.04.2026).

that India should adopt the standards of the Budapest Convention to allow for faster admissibility of digital forensic reports and joint investigation teams. Overall, together, these reforms would shift India from a reactive MLAT-based model to a proactive, rapid-response mechanism to collect evidence in a timely manner and enhance cyber deterrence.

Another stage where the enforcement deficit becomes very conspicuous is the investigation stage. Research from the National Law University Delhi 2023 states that nearly 70% of cyber police stations lack trained professionals in digital forensics and that most of the officers are not able to conduct proper electronic evidence.²⁷ Thus, filing of chargesheets after electronic evidence deteriorates, which results in high complaints but very low rates of conviction. According to the Annual Cybercrime Investigation Report, 2024, prepared by the Data Security Council of India, the average length of a cybercrime investigation in India is about 18-36 months, with conviction rates under 2% primarily due to weak handling of evidence, lack of specialist prosecutors and outdated forensic infrastructure.²⁸ Moreover, reports from training this year indicate that many police stations do not have access to even the most basic digital forensic imaging tools, requiring the refusers to be escalated to already overburdened state labs, where again, months of delay are built into the justice system. Collectively, these points toward a systematic problem, i.e. laws are in place, but enforcement capacity is limited. Regulatory ambiguity creates confusion for platforms on compliance, and limitations in investigative capacity mean that very few cybercrimes that are even reported lead to convictions. Thus, the cyber law problem is not about drafting more laws, but rather about establishing clearer, workable regulatory standards and enforcement institutions with technical capacity.

SUGGESTIONS AND CONCLUSION

As India gears up to frame a regulatory response to the evolving complexities of legal disputes in synthetic media and deepfake systems, it can draw inspiration from three international frameworks that offer insightful lessons.

Firstly, the EU Artificial Intelligence Act adopts a risk-based model, wherein AI is classified as permitted or not permitted based upon the classification and also mandates the labelling of AI-generated or manipulated image, audio or video files (for example, deepfakes) as AI-generated. Thus, the need for mandatory upfront labels and disclosure easily tackles the scope for misuse of

²⁷ National Law University Delhi, “Digital forensics capacity gap in Indian policing” (Research working paper, 2023), *available at*: <https://nludelhi.ac.in> (last visited on: 03.04.2026).

²⁸ Data Security Council of India, “Annual cybercrime investigation report” (2024), *available at*: <https://dsci.in> (last visited on: 03.04.2026).

deepfakes at least at a primary level, given that such regulations will only be practised in legal and regulated intermediaries, giving grounds for an informed social media content community and also places a responsibility on makers and distributors of any such content.

Secondly, in the US, such as the DEEPFAKES Accountability Act and various state laws. Its approach is more focused on individual rights and legal remedies. For instance, creating deepfake content without a person's consent, especially in cases of election manipulation and fake pornography, can be illegal. Victims have the right to approach the court directly and claim compensation.

Thirdly, China's Administration of Deep Synthesis of Internet-based Information Services requires that users must verify their real identity before creating or uploading deepfake content, as it makes it easier to trace offenders. The law required platforms to monitor and control their AI systems and required them to audit and report on how their systems work.

Simply put, India's legal framework on cybersecurity needs to undergo several significant changes in order to keep pace with new technological advancements. Nevertheless, India has transformed its cyberlaws, especially the newly introduced IT Amendment Rules 2026, which explicitly address deepfakes and focus on regulating AI-generated content, but continues to remain reactive and insufficient to address the growing risks posed by deepfakes, as it is majorly intermediary-centric. India can take help from different successful international models, such as risk- based classification from the EU, a victim-centric approach from the US, and user identity verification from China. Ultimately, in order to combat modern cyber threats, India requires a multi-dimensional approach and collaborative efforts